

May 3, 2017

# Operational Considerations for Containers

**Chris Swan**

**VP, CTO Global Delivery Organisation**

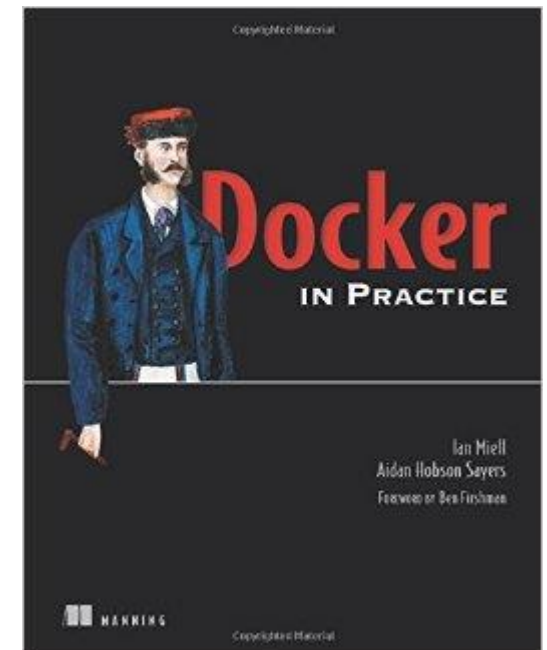
# With thanks to Ian Miell

**‘A Checklist for Docker in the Enterprise’**  
was the inspiration for this talk

<https://zwischenzugs.wordpress.com/2016/07/08/a-checklist-for-docker-in-the-enterprise/>

Shortlink: [is.gd/IMDoCL](https://is.gd/IMDoCL)

**Ian Miell Docker Check List**





# Why me?

Using Docker since v0.53 (summer of 2013).

Built Docker in to a commercial networking platform as its plugin mechanism in early 2014 (with customers in production before v1.0 launch at first DockerCon).

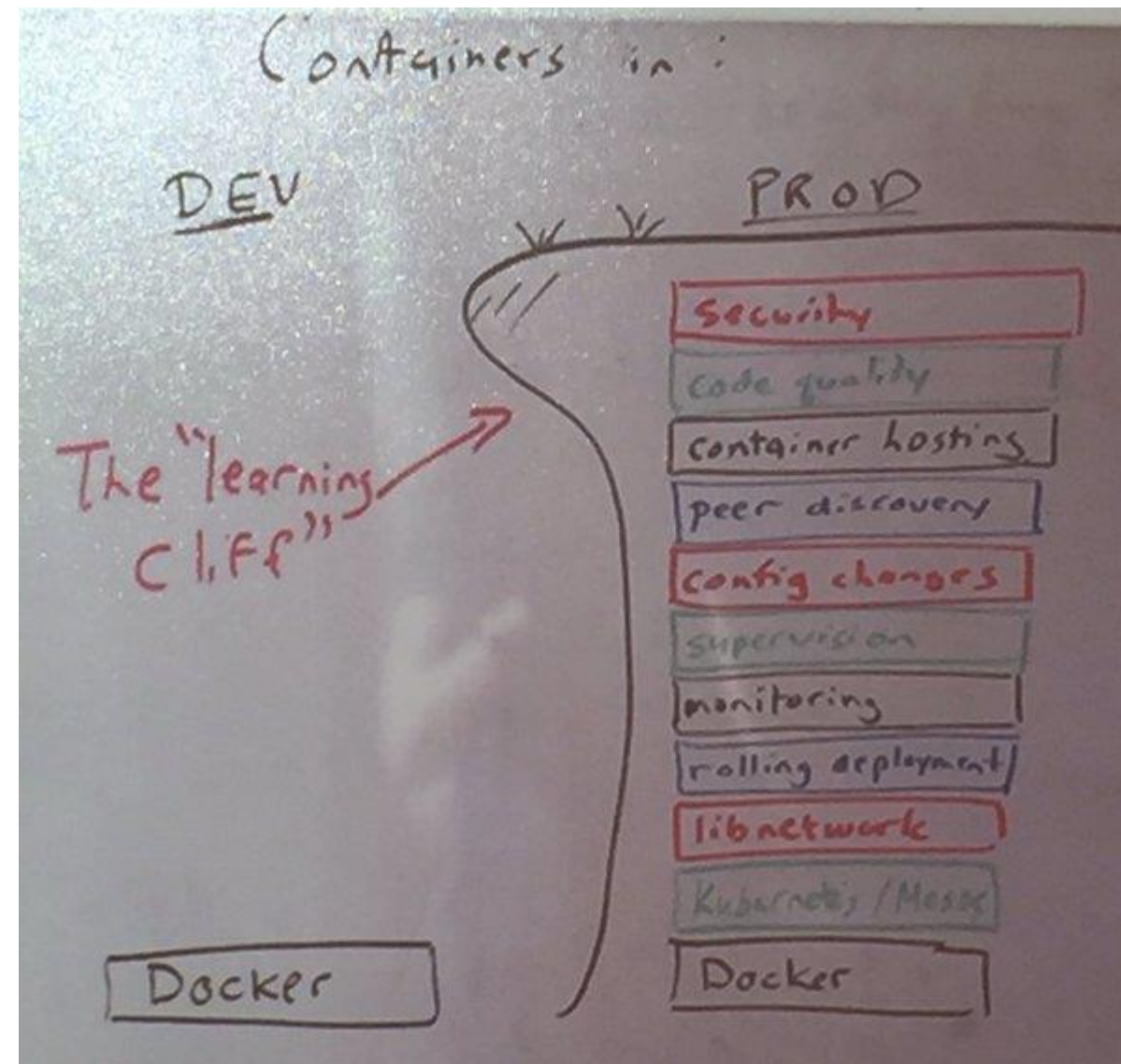
Now using containers to help reskill a 120,000 strong delivery organisation for a world of Infrastructure as Code.

*‘Installation is a software hate crime’*

Pat Kerpan  
(then Borland CTO)  
c. 2004



# The Containers Learning Cliff



Credit: Michael Ducey @mfdii <https://twitter.com/mfdii/status/697532387240996864>

# Images

Registry

Scanning

Image Building

Image Integrity

Base Images

Third Party Images

SDLC

Secrets



*Some (but not all) of this can be handled by Docker Trusted Registry (DTR) or Docker Cloud*

# Operations

Provisioning (Orchestration)

Config Management (Inside and Out)

Monitoring

Logging

*Or use a PaaS?*



# Security and Audit

`docker == root`

Security Monitoring

Forensics

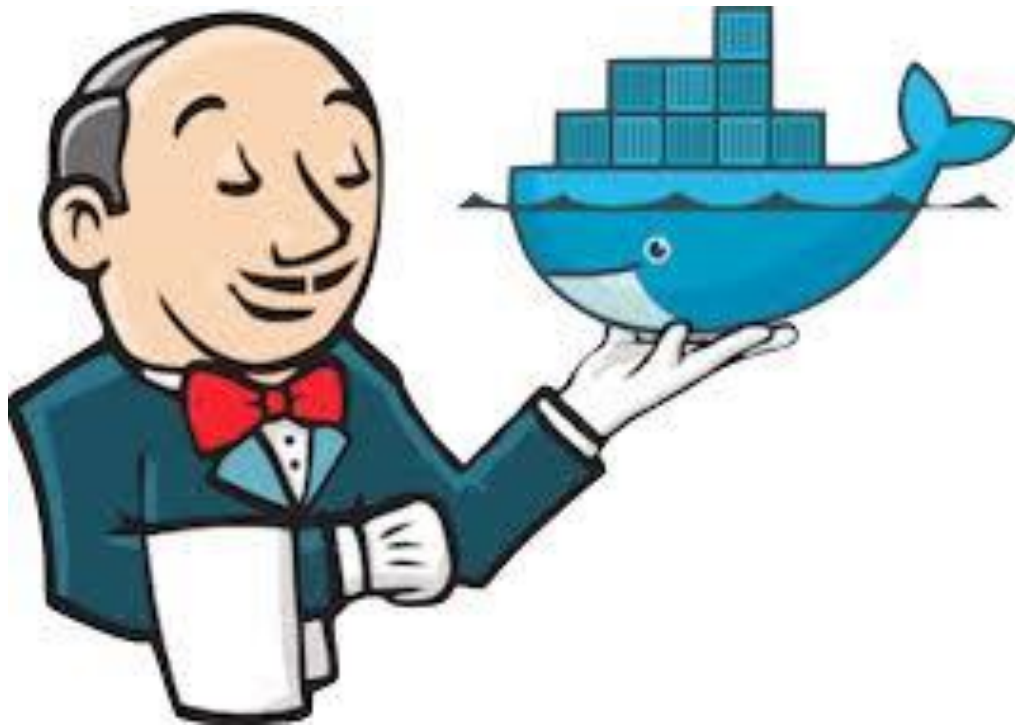


*The CIS Docker Security Benchmark  
has guidance for host config, container config etc.*

*Many aspects can be checked with the DockerBench tool*

# Development

Dev Environments  
CI/CD



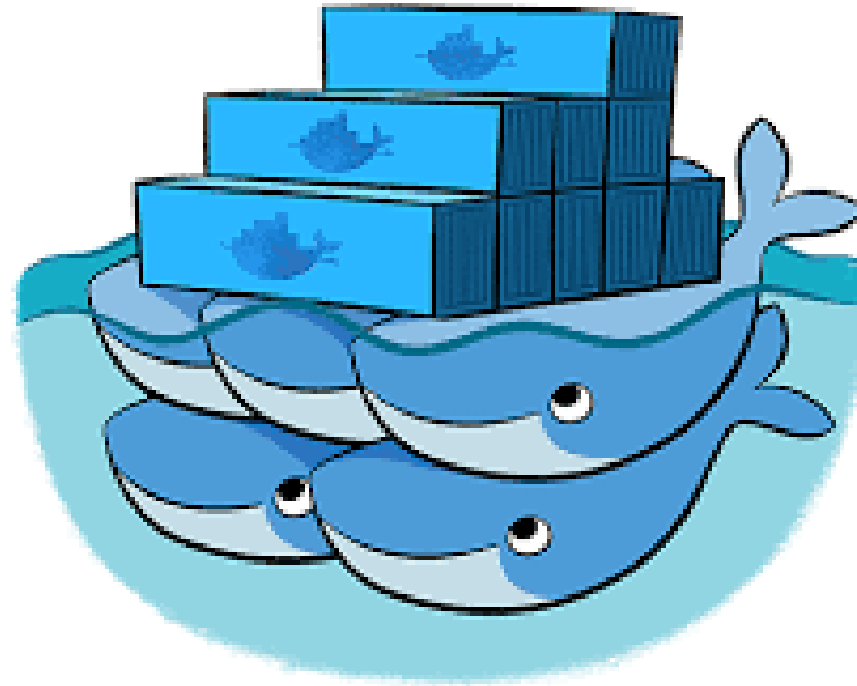
# Infrastructure

Shared Storage

Networking

PaaS

Cloud Providers



# Review

## Images

- Registry
- Scanning
- Image Building
- Image Integrity
- Base Images
- Third Party Images
- SDLC
- Secrets



## Security & Audit

- docker == root
- Monitoring
- Forensics



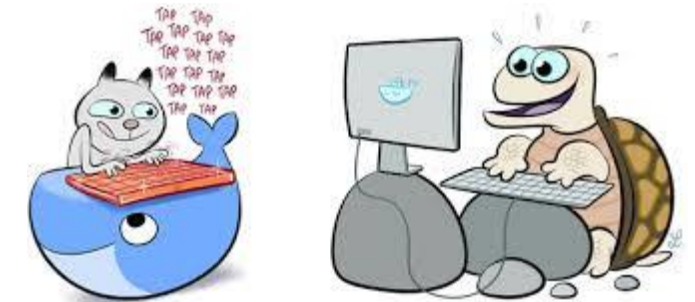
## Operations

- Provisioning
- Config Management
- Monitoring
- Logging



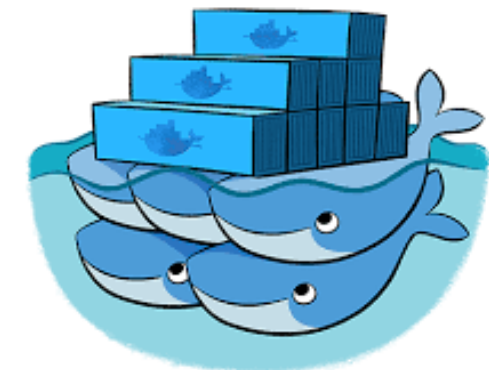
## Development

- Dev Environments
- CI/CD



## Infrastructure

- Shared Storage
- Networking
- PaaS
- Cloud Providers



# Coda - what does ServerLess take away?

## **Images**Libraries

Registry  
Scanning  
Image Building  
Image Integrity  
Base Images  
Third Party Images  
SDLC  
Secrets

## **Security & Audit**

docker == root  
Monitoring  
Forensics

## **Operations**

Provisioning  
**Config Management**  
Monitoring  
Logging

## **Development**

Dev Environments  
CI/CD

## **Infrastructure**

Shared Storage  
Networking  
PaaS  
Cloud Providers



# Questions?